

Computer Science & Information Technology

Theory Of Computation

DPP:1

Regular Expression

Q1 The possible number of DFA with 2 states X,Y over the alphabet {a, b} where X is always initial state ?

Q2 The possible number of DFA with 2 states X,Y over the alphabet {a, b} where X is always initial state , that accepts empty language?

Q3 The possible number of DFA with 2 states X,Y over the alphabet {a, b} where X is always initial state , that accepts complete language?

Q4 Consider the DFA ,M with states $Q = \{0,1,2,3,4\}$, input alphabet $\Sigma = \{0,1\}$ start state 0, final state 0 and transition function $\delta(q, i) = |q^2 - i| \bmod 5$ $q \in Q$, Input alphabets are {0,1}.

The above DFA, M accepts all binary strings containing

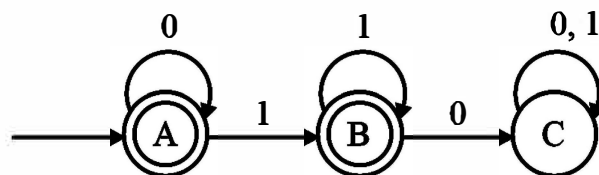
- (A) Even number of 1's
- (B) Odd number of 1's
- (C) Even number of 0's
- (D) Odd number of 0's

Q5 Consider the DFA ,M with states $Q = \{0,1,2,3,4\}$, input alphabet $\Sigma = \{0,1\}$ start state 0, final state 0 and transition function $\delta(q, i) = |q^2 - i| \bmod 5$ $q \in Q$, Input alphabets are {0,1}

The number of states in the minimal finite automata ,which is equivalent to M is

- (A) 1
- (B) 2
- (C) 3
- (D) 4

Q6 The regular expression for the language recognized by the following finite automata is?



- (A) $(0 + 1)^*$
- (B) $0(0 + 1)^*1$
- (C) 0^*1^*
- (D) $0^*(0 + 1)^*1^*$

Q7 Choose the regular language from the following given options

- (A) $L = \{x/x \in (a + b)^*\}$ and x is even length palindrome
- (B) $L = \{a^n / n \geq 1\}$
- (C) $L = \{a^n b^{2n} / n \geq 1\}$
- (D) None

Q8 Which of the following regular expression represents all strings of a's and b's where the length of the string is at most 'n' is

- (A) $(a + b)^n$
- (B) $(a + b)^n (a + b)^*$
- (C) $(a + b + \epsilon)^n$
- (D) None of the above

Q9 Which of the following pair of regular expressions are not equal

- (A) $(r^*)^*$ and $(r^*)^*$
- (B) $(r + \epsilon)^*$ and r^*
- (C) $(rr + \epsilon)^*$ and r^*
- (D) None of the above

Q10 Consider the language S^* , where S is all strings of a's and b's with odd length. The other description of this language is.

- (A) All strings of a's and b's



- (B) All even length strings of a's and b's
- (C) All odd length strings of a's and b's
- (D) None of the above

Q11 Let $r = (1 + 0)^*$, $s = 11^*0$ and $t = 1^*0$ be three regular expressions. Which one of the following

is true?

- (A) $L(s) \subset L(r)$ and $L(s) \subset L(t)$
- (B) $L(r) \subset L(s)$ and $L(s) \subset L(t)$
- (C) $L(t) \subset L(s)$ and $L(s) \subset L(r)$
- (D) None of the above



Answer Key

Q1 64
Q2 20
Q3 20
Q4 (A)
Q5 (B)
Q6 (C)

Q7 (B)
Q8 (C)
Q9 (C)
Q10 (A)
Q11 (A)



Hints & Solutions

Q1 Text Solution:

Total DFA possibilities for final and non-final states (With 2 states DFA) = 4

For Each DFA 16 different DFAs are possible because of transitions.

Total DFA = $4 * 16 = 64$

Q2 Text Solution:

Both the states must be non - finals

For DFA 16 different DFAs are possible because of transitions.

There are 4 different DFAs also exist when Y is final state and Do the reverse transitions.

Total DFAs = $16 + 4 = 20$

Q3 Text Solution:

Total DFAs = 20

Q4 Text Solution:

Correct design of the DFA will be Even number of 1's.

Q5 Text Solution:

The number of states in the minimal finite automata = 2

Q6 Text Solution:

The regular expression = 0^*1^*

Q7 Text Solution:

$L = \{x/x \in (a + b)^*\}$ and x is even length
palindrome = CFL

$L = \{a^n / n \geq 1\} = \text{Regular}$

$L = \{a^n b^{2n} / n \geq 1\} = \text{CFL}$

Q8 Text Solution:

$(a + b)^n$ Produce exactly n length string.

$(a + b)^n (a + b)^*$ Produce atleast n length string.

$(a + b + \epsilon)^n$ Produce atmost n length string.

Q9 Text Solution:

$(rr + \epsilon)^*$ and r^* Both are different because $(rr + \epsilon)^*$ it will generate even length r and r^* can generate all length.

Q10 Text Solution:

S = Odd length

Odd length expression = $(a+b) [(a+b)^2]^*$

$S^* = [(a+b) [(a+b)^2]^*]^*$ is same as $(a+b)^*$ which will generate all the a's and b's.

Q11 Text Solution:

$r = (1 + 0)^*$

$s = 11^*0$

$t = 1^*0$

$L(s) \subset L(r)$ and $L(s) \subset L(t)$

Therefore, Option (a) is correct.

